

For example, gcc can compile the code for ARM7, ARM9, MSP430, AVR, etc. Therefore code written for, say, MSP430, which has been compiled by a gcc, can be easily compiled by gcc for AVR as well. The only thing that needs to be changed is hardware-dependant modules that deal explicitly with the peripheral registers. This also makes the code future-proof to changes in the hardware at a later stage," explains Jain. Since the source code for these tools is open source, it is likely that the same tool can be ported from one platform to other, thus keeping the usage of tools common across multiple platforms.

"Open source enables to address technology evaluation and proof-of-concepts faster, as most of the tools are available for download from the Internet and can run on standard hardware configurations," shares Nehul Gullaiya, design engineer, embedded group, KritiKal Solutions.

We all are well aware of the support and activities of open source community. "The number of users in open source community is extremely large, contributing their experience to get the tool tested in umpteen number of environments and thus making it more stable," shares Walia.

"All the loopholes, problems and limitations are openly discussed and therefore the overall community gets faster solution. In most of the cases, open source tools help in cost saving, standardisation and faster turnaround time for project execution," adds Walia.

OSS tools—benefits to the organisation

As there is no licensing cost associated with open source tools, organisations save a big deal of money if they wisely use open source tools. Walia shares, "A design house has to continuously work on lots of different kinds of designs involving various process architectures, etc. If the organisation has to buy proprietary tools for each of the function, then the investments will become very high. In such cases, open source tools

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become very handy as they come free."

Another advantage of open source tools is that you can standardise the same set of tools across multiple platforms. "The knowledge of usage on one platform helps us in not duplicating the same learning effort in multiple platforms," comments Walia.

Nair says, "Choosing between open source and proprietary is a trade-off between costs and convenience/productivity. If your immediate revenue is low, or non-existent—as is the case with most start-ups—then go for open source tools if a good port exists. You may not squeeze the last clock-cycle, or the last byte, but can certainly get away by writing assembly for little parts."

Since there are so many people across the globe, a popular open source tool gets tested by many on different platforms. This experience can help an organisation in making informed decision. However, it holds good only for popular tools.

gcc and gdb

"GNU compiler collection (gcc) and GNU debugger (gdb) are the most popular open source tools, primarily because these are available for a wide range of processors and microcontrollers," says Jain.

Nair adds, "The gcc is not a generic compiler. It can more appropriately be characterised as a compiler generator for many platforms." Thus you have gcc ARM, avr-gcc, etc.

Hardik Shah, an expert in embedded systems at Think Labs, shares, "The experience says that gcc compilers are very powerful. Also, for debugging, gdb is a very powerful command

line debugging tool. Insight, which is a GUI-based debugging tool, uses gdb at backend. The combination of Insight and gdb is used as a debugging tool for many targets."

A gcc compiler can be ported to many different architectures. "However, embedded compilers built for specific targets can be far more efficient. Some of the compilers are lagging behind in terms of technology," argues Chris Hills, technical specialist, Phaedrus Systems.

Nair adds, "Since gcc is not an engineered product, there is a lot of redundant/dead source code. However, there is no surety that the proprietary source has done its cleaning up to the optimal level."

Eric Bressie, principal software engineer from Transition, says, "gcc has worked towards moving some of the duplicate code into shared code (such as the libgloss portion of gcc), but this takes time. Since there are so many architectures, languages and sub-features included, which can be developed independently, it is difficult to have them coordinated and ensure regression issues do not come about."

The Eclipse

The Eclipse integrated development environment (IDE) has become a standard for embedded development. "Eclipse IDE can run on any operating system since it is a Java-based application which requires a Java runtime environment. Some processor makers provide an Eclipse-based tool chain, like ARM has created its own workbench tools such as open source Eclipse 3.3 IDE," shares Shah.

There are many developers now contributing to create the plugins for the target platforms. One would require a tool chain specific to the target processor and built as a plug-in, or as an Eclipse RCP application.

Shah explains, "You can download and install the open source cross-development tools such as GNU-ARM or WINARM toolchains in your OS, and you will just need to install the plugins